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ORIGINAL RESEARCH

Out-of-hospital management of unresponsive, apneic, witnessed opioid overdoses: a case series from a supervised consumption site

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Abstract

Objectives There are conflicting recommendations for lay rescuer management of patients who are unresponsive and apneic due to opioid overdose. We evaluated the management of such patients at an urban supervised consumption site.

Methods At a single urban supervised consumption site in Vancouver, BC, we conducted a retrospective chart review and administrative database linkage of consecutive patients who were unresponsive and apneic following witnessed opioid overdose between January 1, 2012 and December 31, 2017. We linked these visits with regional hospital records to define the entire care episode, which concluded when the patient was discharged from the supervised consumption site, ED, or hospital, or died. The primary outcome was successful resuscitation, defined as alive and neurologically intact (ambulatory and speaking coherently, or alert and oriented, or Glasgow Coma Scale 15) at the conclusion of the care episode. Secondary outcomes included mortality and predefined complications of resuscitation.

Results We collected 767 patients, with a median age of 43 and 81.6% male, with complete follow-up on 763 patients (99.5%). All patients were managed with oxygen and ventilation (100%, 95% CI 0.995–1.0); 715 (93.2%, 95% CI 0.911–0.949) received naloxone; no patients underwent chest compressions (0%, 95% CI 0–0.005). All patients with complete follow-up were alive and neurologically intact at the end of their care episode (100%, 95% CI 0.994–1.0). Overall, 191 (24.9%) patients were transported to hospital, and 15 (2.0%) patients required additional naloxone after leaving the supervised consumption site; 16 (2.1%) developed complications, and 1 patient was admitted to hospital.

Conclusions At an urban supervised consumption site, all unresponsive, apneic patients with witnessed opioid overdose were successfully resuscitated with oxygen and/or naloxone. No patients required chest compressions.

Keywords Opiate overdose · Harm reduction · Ventilation · Cardiopulmonary resuscitation

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Résumé

Objectifs Il existe des recommandations contradictoires concernant la prise en charge par des secouristes non professionnels des patients qui ne réagissent pas et sont apnéiques en raison d'une surdose d'opioïdes. Nous avons évalué la prise en charge de ces patients dans un site urbain de consommation supervisée.

Méthodes Dans un seul site de consommation supervisée urbain à Vancouver, en Colombie-Britannique, nous avons effectué un examen rétrospectif des dossiers et un couplage de bases de données administratives de patients consécutifs qui étaient insensibles et apnéiques après avoir été témoins d'une surdose d'opioïdes entre le 1er janvier 2012 et le 31 décembre 2017. Le résultat primaire était la réussite de la réanimation, définie comme étant vivante et neurologiquement intacte (ambulatoire et parlant de manière cohérente, ou alerte et orientée, ou échelle de coma de Glasgow 15) à la fin de l'épisode de soins. Les résultats secondaires comprenaient la mortalité et les complications prédéfinies de la réanimation.

Résultats Nous avons recueilli 767 patients, avec un âge médian de 43 ans et 81,6 % d'hommes, avec un suivi complet de 763 patients (99,5 %). Tous les patients ont été pris en charge avec de l'oxygène et la ventilation (100 %, IC à 95 % : 0,995-1,0) ; 715 (93,2 %, IC à 95 % : 0,911-0,949) ont reçu de la naloxone ; aucun patient n'a subi de compressions thoraciques (0 %, IC à 95 % : 0-0,005). Tous les patients ayant fait l'objet d'un suivi complet étaient vivants et intacts sur le plan neurologique à la fin de leur épisode de soins (100 %, IC à 95 % : 0,994-1,0). Dans l'ensemble, 191 (24,9 %) patients ont été transportés à l'hôpital, et 15 (2,0 %) patients ont eu besoin de naloxone supplémentaire après avoir quitté le site de consommation supervisée ; 16 (2,1 %) ont développé des complications, et 1 patient a été admis à l'hôpital.

Conclusions Dans un centre de consommation supervisée urbain, tous les patients apnéiques non réceptifs ayant été témoins d'une surdose d'opioïdes ont été réanimés avec succès avec de l'oxygène et/ou de la naloxone. Aucun patient n'a eu besoin de compressions thoraciques.

Clinician's capsule

What is known about the topic?
There are conflicting recommendations for management of opioid overdose causing unresponsiveness and apnea in the community.

What did this study ask?
In unresponsive, apneic patients with witnessed opioid overdose at a supervised consumption site, can ventilation and naloxone successfully resuscitate patients?

What did this study find?
All patients were treated with ventilation, and all patients with complete follow-up were neurologically intact.

Why does this study matter to clinicians?
Ventilation and naloxone administration are appropriate initial steps for trained providers to manage witnessed apneic and unresponsive opioid overdoses.

Introduction

Death caused by opioid overdose is a growing, devastating public health crisis, with over 17,000 unintentional deaths recorded in Canada between 2016 and 2020 [1]. With the vast majority of deaths occurring outside of hospital [2], clear, evidence-based training for lay rescuers responding to opioid overdoses is essential to prevent avoidable deaths.

Considering the pathophysiology of opioid toxicity [3], it is expected that overdose will cause respiratory arrest before cardiac arrest. Unfortunately, laypersons may have difficulty accurately detecting a pulse [4, 5] which makes differentiating respiratory arrest from cardiac arrest challenging. Due to limited evidence evaluating management and outcomes of patients experiencing opioid overdoses in the community, current guidelines provide conflicting recommendations [6–15]. The World Health Organization and harm reduction organizations have recommended airway management, mouth-to-mouth ventilation, and naloxone administration when managing an unresponsive patient with opioid overdose [7]. However, the International Liaison Committee on Resuscitation guidelines for suspected opioid poisoning recommend untrained laypersons administer chest compressions and naloxone without checking a pulse when assisting unresponsive and apneic patients with opioid overdose [8]. Applying the same management algorithm to all unresponsive opioid overdoses risks inappropriately administering chest compressions to patients who are in respiratory arrest but not cardiac arrest.

Therefore, in patients who are apneic and unresponsive after a witnessed opioid overdose in a community setting, initial management—ventilation versus chest compressions—remains uncertain. Witnessed opioid overdoses with rapid intervention are an important subset of overdoses that may benefit from ventilation alone, obviating the need for potentially harmful chest compressions [16–19]. We evaluated whether ventilation and naloxone provide effective resuscitation of unresponsive, apneic patients with witnessed opioid overdose in a supervised consumption setting.

Methods

Study design

We performed a retrospective chart review and administrative database linkage of adult patients at least 16 years old with opioid overdose at a supervised consumption site is called Insite in Vancouver, BC.

Study setting

Insite consists of 13 injection stalls where participants use pre-obtained drugs via route of their choosing under the observation of Insite staff. A monitored post-consumption lounge is adjacent to the consumption room. Insite employs RNs, licensed practical nurses, and peer responders, all of whom are trained in overdose response [20]. This includes naloxone administration, oxygen administration, non-invasive ventilation with bag-valve-mask, and chest compressions. Online medical oversight is available. In 2017, Insite had an average of 356 visits per day [21]. Participants register with a unique identifier termed a "handle" that is used for all subsequent visits, and may choose to provide identifying information such as their name, date of birth and gender.

In cases of presumed opioid overdose, the patient is immediately assessed by staff, who perform all interventions, including EMS activation, at their discretion. Any opioid overdose requiring intervention is documented in a chart linked to the participant's handle in Insite's database. This includes date, time, and location of the overdose, likely culprit substance, vital signs, and all interventions with patient responses.

Population

We screened all Insite charts that were coded as 'opioid overdose' to obtain consecutive cases between January 1, 2012 and December 31, 2017. We included patients meeting the following criteria: (1) unresponsive, defined as Glasgow Coma Scale of 3, or unresponsiveness to all physical and verbal stimulation before administration of any resuscitative intervention; and (2) respiratory arrest, defined as initial respiratory rate of zero breaths per minute, apnea, agonal breathing, or central cyanosis. Documented presence or absence of a pulse was not required for study inclusion. We excluded patients whose overdose occurred outside of Insite, or patients whose symptoms were inconsistent with opioid overdose, such as agitation. Patients could be included more than once, and we assumed that each discrete episode was independent.

Data collection

We adhered to published guidelines for retrospective chart review studies [22]. Trained abstractors independently extracted data from charts using a standardized collection instrument in REDCap [23, 24]. Abstractors were aware of the study hypothesis but were blinded to all linked outcomes. Abstractors coded data using a priori definitions and fields included demographics; initial vital signs; resuscitation interventions including painful stimuli, naloxone administration, ventilatory support (oxygen, non-invasive ventilation with bag-valve-mask, extra-glottic device insertion, or intubation), chest compressions; length of resuscitation; and outcomes including neurological status, final vital signs, and disposition. The latter could include ambulatory discharge, transfer to hospital via ambulance, or death on-scene (Online resource Appendix A for definitions). We defined cardiac arrest as absence of a pulse and initiation of chest compressions by a health care professional. We could not include missing data, but conflicting data were resolved at regularly scheduled meetings. For cases where missing data limited the abstractor's ability to determine whether the patient met inclusion criteria or if important outcomes (type of resuscitative intervention, mortality, or neurologic outcome) were coded as indeterminate, the chart was flagged and a second assessor reviewed the chart to verify. Data for cases with insufficient information to determine whether the patient met inclusion criteria (i.e. insufficient description of level of consciousness or respirations) were analyzed as a separate cohort to assess for systemic differences from patients included in the study. To assess inter-rater reliability, we reviewed a random sample of 10% of charts and calculated a Cohen's kappa value based upon EMS attendance.

For patients transferred to hospital, we identified the related ED visit by linking the Insite database with the regional ED database using their provincial personal health number. Patients were identified using the participant's name provided to Insite (if a name was provided), which was cross-referenced to the Vancouver Coastal Health authority primary care electronic medical record to identify their personal health number. When available, we used gender and date of birth from Insite records as secondary identifiers to support the match to a personal health number. The date and time of the initial Insite encounter and the patient's personal health number were used to link Insite visits to ED visits.

For patients transferred to hospital, AR reviewed ED and hospital charts to obtain initial vital signs, resuscitative interventions, and final disposition at the end of their care episode: discharge from hospital, admission to hospital, or death. All data were recorded in a standardized data collection instrument in REDCap [23, 24]. In cases where patients who were not neurologically intact on-scene and had transport to hospital initiated, but no subsequent linked

ED visit could be identified, we recorded available information on their neurological status at time of Insite disposition. Furthermore, for such patients, we accessed Insite and ED records from the two closest hospitals using the patient handle or personal health number to see if they accessed care at Insite or the ED up until June 21, 2021.

Outcome measures

The a priori primary outcome was intact neurological function—defined as ambulating independently and conversing normally, alert and oriented, or Glasgow Coma Scale 15—at the conclusion of the care episode. If patients were transported from Insite and we were unable to link the patient to an ED record, but the patient had a repeat Insite or ED visit, we considered the patient neurologically intact at the conclusion of the care episode. Secondary outcomes included mortality, cardiac arrest, naloxone administration, and predefined complications of resuscitation (Online resource Appendix A).

Data analysis

We analyzed data using STATA SE 15.1 (Stata Corp, College Station, TX). We described categorical variables as proportions with 95% CIs, and continuous variables as medians with interquartile ranges. Some patients had insufficient data to ascertain whether their episode involved an unresponsive witnessed opioid overdose, so we obtained demographics, interventions, and outcomes to evaluate for systematic differences when compared with included patients.

Ethics

The UBC Clinical Research Ethics Board, and the Vancouver Coastal and Providence health authorities approved this study.

Results

Study population

We identified 5875 cases coded as 'opioid overdose' between January 1, 2012 and December 31, 2017. We excluded 364 cases (6.2%) due to missing information that limited our ability to determine if they met inclusion criteria. Demographics, resuscitative interventions, and neurological status among patients with missing data were similar to included patients (Online resource Appendix B). Overall, we analyzed 767 patients (Fig. 1) with 81.6% male and a median age of 43 years. The kappa value for EMS attendance was 0.87.

Characteristics of resuscitation

All 767 patients received oxygen and/or non-invasive ventilation (100%, 95% CI 0.995–1.0). No patients received chest compressions (0%, 95% CI 0–0.005). Naloxone was administered to 715 patients (93.2%, 95% CI 0.911–0.949). Median naloxone dose and route are presented in online resource Appendix C. Methods and duration of non-invasive ventilation are described in Table 1. No patients received airway support by intubation, extra-glottic device, or mouth-to-mouth ventilation.

Post-resuscitation status

All 767 patients were alive at time of disposition from Insite (Table 2) with 675 (88.0%) recorded as neurologically intact at that time (Table 3). This left 92 patients, 44 of whom were transported to hospital and were neurologically intact at the time of ED or hospital discharge. The 48 remaining patients were discharged from Insite with insufficient information recorded to ascertain neurologic status, although 10 were ambulatory, 22 were speaking with no description of coherency, and 16 were breathing and moving spontaneously; 44 of these patients had subsequent Insite or ED visits. Clinical vignettes for the four patients with no identified subsequent Insite or ED visits are in Online resource Appendix D. Therefore, of the 763 patients with complete follow-up, all were neurologically intact at the end of their care episode (100%, 95% CI 0.994–1.0).

For the 364 patients excluded due to insufficient information, 289 (79.4%) were recorded as neurologically intact by the end of their Insite care episode. The remaining 75 were discharged from Insite with insufficient information to ascertain neurologic status; 72 of these patients had a subsequent Insite or ED visit. Clinical vignettes for the three patients with no identified subsequent Insite or ED visits are in Online resource Appendix D.

Disposition

Insite staff transferred care of 191/767 patients (24.9%) to paramedics (Table 2), and we identified 115 linked ED visits. Fifteen patients (2.0%) required additional naloxone after leaving the supervised consumption site, 114 patients were discharged directly from the ED, and 1 patient was admitted to hospital for less than 24 h; all patients were neurologically intact at discharge.

Complications

Sixteen patients (2.1%) had documented post-resuscitation complications (Table 4), with one severe complication (aspiration pneumonia requiring hospitalization).

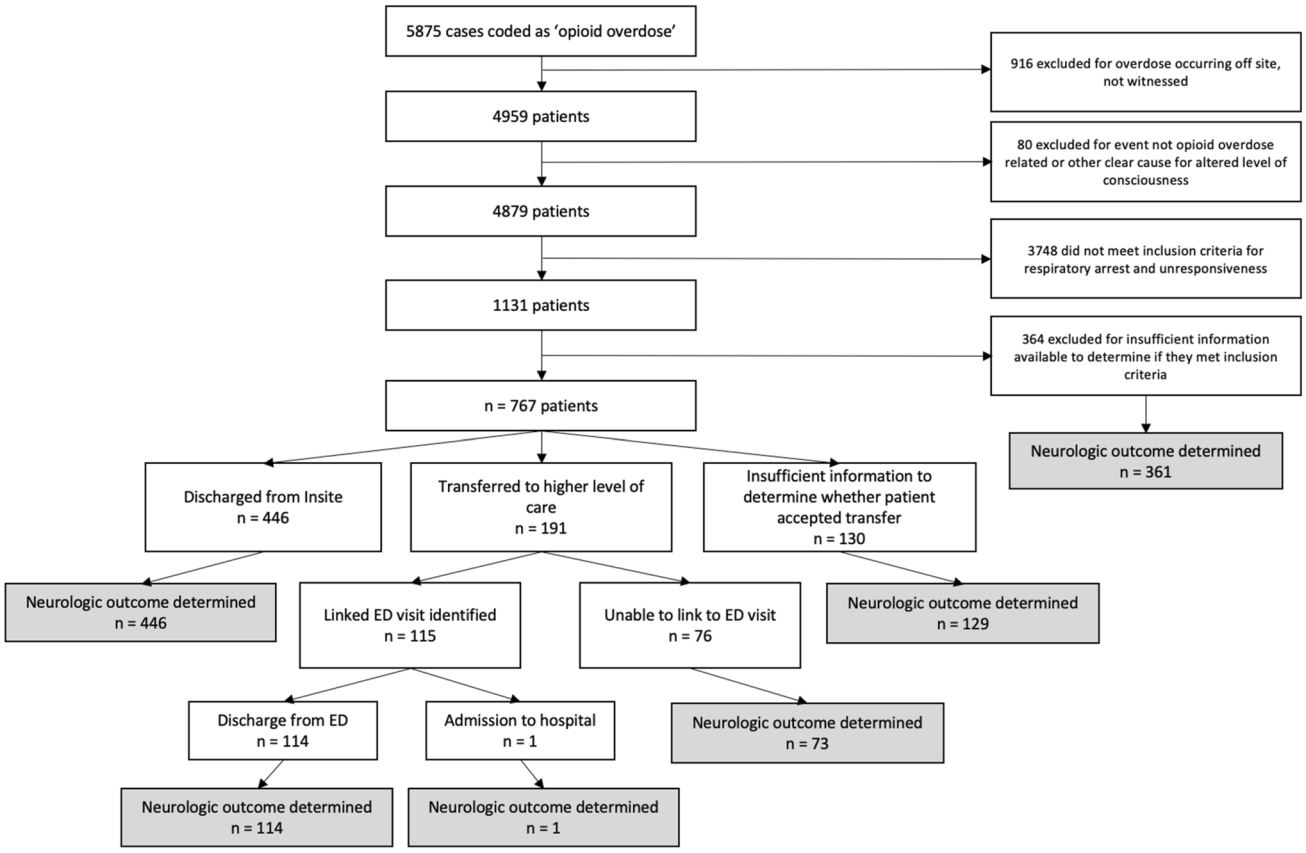

Fig. 1 Identification and exclusion of patients in study population

Discussion

Interpretation

Our findings demonstrate that supervised consumption sites are a life-saving harm reduction tool. Over a 6 year period at an urban supervised consumption site, all witnessed, apneic and unresponsive opioid overdoses were managed with non-invasive ventilation and oxygen, with none requiring chest compressions. All patients with complete follow-up were discharged alive and neurologically intact from their care episode. Our analysis of patients excluded due to incomplete data demonstrates similar demographics, treatments, and outcomes, and provides confidence that such patients were not systematically different from included patients. Our cohort was over 80% male, similar to the population that attends the supervised consumption site [21], prior ED studies [25], and provincial opioid overdose data [26]. Our data cover the emergence of ultra-potent opioids to the Canadian illicit drug market in 2014 [27]. The introduction of ultra-potent opioids did not change the nature of resuscitative interventions administered in our cohort, though other studies suggest they may increase the cumulative dose of naloxone required for reversal [28]. Our findings illustrate that ventilation and naloxone can successfully resuscitate patients who are comatose and apneic due to a witnessed opioid overdose. Furthermore, the complication rate appears to be reassuringly low.

Previous studies

To our knowledge, there are no other studies evaluating resuscitation interventions and outcomes in a supervised consumption site setting.

Clinical implications

Our data support the effectiveness of non-invasive ventilation by trained healthcare workers and peer responders for treating opioid overdose. For laypersons, mouth-to-mouth ventilation may be an effective ventilation strategy [29, 30]. Despite assumptions that rescuers are unwilling to provide mouth-to-mouth ventilation, surveys of bystanders attending cardiac arrests have found only 1% objected to performing mouth-to-mouth ventilation [31]. Rescuers may be less willing to provide mouth-to-mouth ventilation to strangers [32], but since many overdoses occur in private residences or community housing [2], rescuers may recognize the patient

Table 1 Patient characteristics and resuscitation methods

	Patients (%) $n=767$
Age (years)	
Median (range), [IQR]	43 (15–74), [33–53]
Age group	
15–19	5 (0.7)
20–29	154 (20.1)
30–39	223 (29)
40–49	157 (20.5)
50–59	151 (19.7)
≥60	77 (10)
Self-reported gender	
Male	626 (81.6)
Transgender	4 (0.5)
Female	129 (16.8)
Unknown	8 (1)
Method of resuscitation	
Ventilatory support only	767 (100)
Chest compressions only	0 (0)
Ventilatory support + chest compressions	0 (0)
Unknown	0 (0)
Manual pulse check performed	
Pulse detected	216 (28.1)
No pulse detected	0 (0)
Pulse check not documented	551 (71.8)
Naloxone administration	
Yes	715 (93.2)
No	52 (6.8)
Unknown	0 (0)
Method of oxygen supplementation	
Intubation	0 (0)
Extra-glottic device	0 (0)
Bag-valve mask	659 (85.9)
Face mask/non-rebreather	190 (24.7)
Mouth-to-mouth	0 (0)
Type of airway adjunct used	
Oropharyngeal airway	375 (48.9)
Nasopharyngeal airway	127 (16.6)
Oropharyngeal and nasopharyngeal airways	79 (10.3)
None	82 (10.7)
Unknown	104 (13.5)
Duration of resuscitation (minutes)	$n=217$
Median [IQR]	10 [7–15]
Range	0–130

Table 2 Patient disposition from insite

	Patients (%)
Mortality	$n=767$
Alive	767 (100)
Dead	0 (0)
Transport to hospital	$n=767$
Transported to hospital	191 (24.9)
Declined transport to hospital	217 (28.3)
Not offered transport to hospital	229 (29.9)
Insufficient information	130 (17)
Course in hospital	$n=115$
Neurologically intact on arrival	98 (85.2)
Received naloxone after leaving Insite	
Yes	15 (13)
No	98 (85.2)
Unknown	2 (1.7)
Received oxygen or non-invasive ventilation in hospital	1 (0.8)
Discharge from hospital	
Home from emergency department	86 (74.8)
Admitted to ward	1 (0.9)
Left against medical advice	28 (24.3)
Left hospital neurologically intact	115 (100)

Table 3 Neurologic outcome post-resuscitation

	Patients (%) $n=767$
Neurologically intact at discharge from Insite	675 (88.0)
Not neurologically intact at discharge from Insite	13 (1.7)
Neurologically intact in hospital	10/13
No linked hospital record	3/13
Repeat ED or Insite visit	2/3
Insufficient information to ascertain if neurologically intact at discharge from Insite	79 (10.3)
Neurologically intact in hospital	34/79
No linked hospital record	45/79
Repeat ED or Insite visit	41/45

and be willing to provide mouth-to-mouth ventilation. Many take-home naloxone kits contain barrier breathing devices, which may further mitigate hesitancy. Data from take-home naloxone kit information forms suggest peer responders are willing to provide mouth-to-mouth ventilation, as 46.9% of respondents reported administering mouth-to-mouth ventilation [33]. Studies have found it is possible to generate effective tidal volumes [29] and maintain an oxygen saturation >90% with mouth-to-mouth ventilation [30], although confirmation is required in patients with opioid overdose.

Chest compressions may result in unnecessary trauma in patients with an overdose who are in respiratory but not cardiac arrest. The incidence of rib fractures may vary from 12.9 to 96.6% [16], and other serious injuries such as pneumothorax, gastric perforation, and cardiac contusions have been reported [16–19].

Table 4 Complications of resuscitation

	Recorded at the supervised consumption site (%) $n=767$	Recorded in hospital (%) $n=115$
Minor complications	6 (0.8)	11 (9.6)
Emesis	4 (0.5)	2 (1.7)
Dyspnea—no acute cause identified	1 (0.1)	3 (2.6)
Epistaxis	1 (0.1)	0 (0)
Chest pain—no acute cause identified	0 (0)	6 (5.2)
Rib fracture	0 (0)	0 (0)
Sternal fracture	0 (0)	0 (0)
Pneumothorax—no intervention required	0 (0)	0 (0)
Hemothorax—no intervention required	0 (0)	0 (0)
Severe complications	0 (0)	1 (0.9)
Aspiration pneumonia requiring admission	0 (0)	1 (0.9)
Pneumothorax—intervention required	0 (0)	0 (0)
Hemothorax—intervention required	0 (0)	0 (0)
Hollow viscus injury	0 (0)	0 (0)
No complications documented	761 (99.2)	103 (89.6)

NB: A total of 16 patients had complications recorded, as 2 patients had complications documented at Insite and in hospital

Research implications

While our findings are encouraging, it is important to ascertain whether lay rescuers can successfully resuscitate unresponsive apneic patients without chest compressions. Furthermore, since the majority of opioid overdoses do not take place in a supervised consumption site setting, our findings should be evaluated elsewhere.

Limitations

This study was conducted at a single supervised consumption site where nurses and peer workers have substantial experience in managing opioid overdose. No patients received mouth-to-mouth ventilation, and our findings cannot be extrapolated to a lay rescuer setting, nor can they be applied to unwitnessed overdoses. Ideally, our findings could be compared to a cohort of similar patients who underwent different resuscitative interventions (i.e. received chest compressions or resuscitated by lay rescuer), but we did not have data on such patients. We selected our study population by reviewing charts coded as 'opioid overdose', so it is possible that we missed inaccurately coded episodes. It is possible that incorrect documentation led to the inclusion of inappropriate patients, although the magnitude and direction of bias are unknown. While we describe management of conflicting data, we could not ascertain missing data and there may be a systematic undercount of interventions (for example, 6.8% of patients were not recorded as having received naloxone, and we cannot confirm whether this is true). It is also theoretically possible that chest compressions were not recorded. We were unable to use a validated neurological function score for our primary outcome due to limited data recorded in Insite charts. We cannot be certain that patients did not sustain minor hypoxic injuries related to their overdose. Our method of linking personal health numbers with Insite clients is not validated; nonetheless our use of multiple variables to determine this association improves the robustness of our method. Our administrative data linkages and medical record review were unable to ascertain neurologic recovery in four patients, although all had some component of recovery. Finally, we cannot guard against the possibility of delayed presentation of complications.

Conclusion

Witnessed opioid overdoses causing respiratory arrest were effectively managed by non-invasive ventilation and naloxone administration in a supervised consumption site setting. Future research evaluating the effectiveness of lay rescuer administration of mouth-to-mouth ventilation in witnessed opioid overdose is needed.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s43678-022-00326-9.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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